

ADVANCED DATA ANALYSIS & BUSINESS INTELLIGENCE

DAX Measures Documentation

Superstore Sales & Profit Performance Analysis

Project: Week 3 Advanced Data Analysis & Business Intelligence Dashboard

Dataset: Superstore Sales Dataset

Tool: Microsoft Power BI

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1. Purpose of this Document

This document records the Data Analysis Expressions (DAX) measures developed for the Week 3 Superstore Advanced Data Analysis and Business Intelligence Dashboard. The measures were created to support KPI reporting, profitability analysis, customer analysis, order analysis and interactive dashboard visualisations.

2. Measures Overview

A total of eight DAX measures were created, exceeding the minimum requirement of five calculated measures. Each measure is documented below with its formula, calculation logic and business purpose.

1. Total Sales

DAX Formula

Total Sales = SUM(Table2[Sales])

Calculation Logic: Adds all values in the Sales column to calculate total revenue.

Business Purpose: Used to measure overall sales performance and compare revenue across time periods, regions, categories, segments and products.

2. Total Profit

DAX Formula

Total Profit = SUM(Table2[Profit])

Calculation Logic: Adds all values in the Profit column to calculate overall profit.

Business Purpose: Used to assess profitability across different areas of the business and identify weak or loss-making performance.

3. Total Orders

DAX Formula

Total Orders = DISTINCTCOUNT(Table2[Order ID])

Calculation Logic: Counts each unique Order ID once, regardless of the number of products included in the order.

Business Purpose: Used to measure the volume of customer orders and to support average sales per order analysis.

4. Total Customers

DAX Formula

Total Customers = DISTINCTCOUNT(Table2[Customer ID])

Calculation Logic: Counts each unique Customer ID once.

Business Purpose: Used to measure the size of the active customer base and support customer performance analysis.

5. Profit Margin

DAX Formula

Profit Margin = DIVIDE([Total Profit], [Total Sales], 0)

Calculation Logic: Divides Total Profit by Total Sales. The third argument returns zero where division cannot be performed.

Business Purpose: Used to assess how efficiently sales are converted into profit and to compare profitability across business areas.

6. Average Sales per Order

DAX Formula

Average Sales per Order = DIVIDE([Total Sales], [Total Orders], 0)

Calculation Logic: Divides total sales by the number of unique orders.

Business Purpose: Used to measure the average revenue generated per customer order.

7. Average Discount

DAX Formula

Average Discount = AVERAGE(Table2[Discount])

Calculation Logic: Calculates the arithmetic average of the values recorded in the Discount column.

Business Purpose: Used to monitor the level of discounting and support the analysis of the relationship between discounts and profitability.

8. Loss-Making Orders

DAX Formula

Loss-Making Orders = CALCULATE(DISTINCTCOUNT(Table2[Order ID]), FILTER(Table2, Table2[Profit] < 0))

Calculation Logic: Filters the dataset to rows with negative profit and then counts the distinct Order IDs within the filtered result.

Business Purpose: Used to identify and monitor orders associated with negative profitability.

3. How the Measures Support the Dashboard

The DAX measures provide a consistent calculation layer for the Power BI dashboard. Rather than relying only on automatic aggregation, the measures allow business performance to be calculated consistently across different visuals and filter selections.

- Total Sales and Total Profit support the main financial KPIs and comparative analysis.
- Total Orders and Total Customers provide operational and customer-level performance measures.
- Profit Margin supports profitability analysis beyond absolute sales and profit values.
- Average Sales per Order provides an indicator of average transaction value.
- Average Discount supports the investigation of discounting as a potential driver of reduced profitability.
- Loss-Making Orders supports the identification of negative profitability and areas requiring further investigation.

4. Conclusion

The eight DAX measures developed for this project provide the calculation foundation for the Week 3 Power BI dashboard. Together, they support analysis of revenue, profitability, orders, customers, transaction value, discounting and loss-making activity. The measures also enable interactive analysis because their results respond to filters and slicers applied within the Power BI report.